

Piecewise Linear Regression

Optimization problem:

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

with data pairs $\{(a_i, b_i)\}_{i=1}^n$.

Equivalent linear programming:

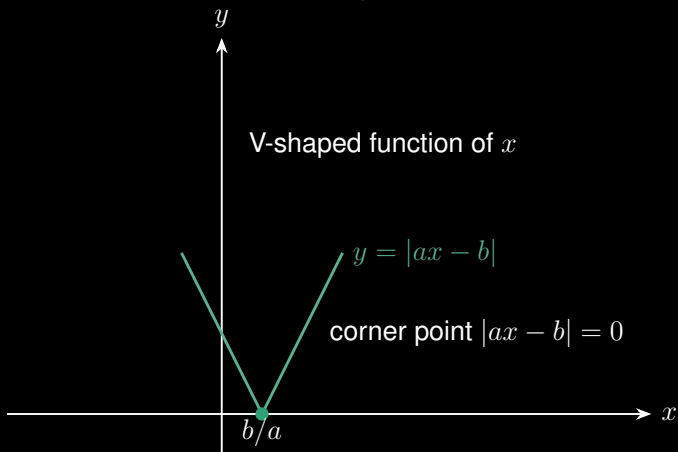
$$\begin{aligned} \min_{x, t_1, \dots, t_n} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & -t_i \leq a_i x - b_i \leq t_i, \quad \forall i \end{aligned}$$

Piecewise Linear Regression

How to minimize $\sum_{i=1}^n |a_i x - b_i|$ without using linear programming?

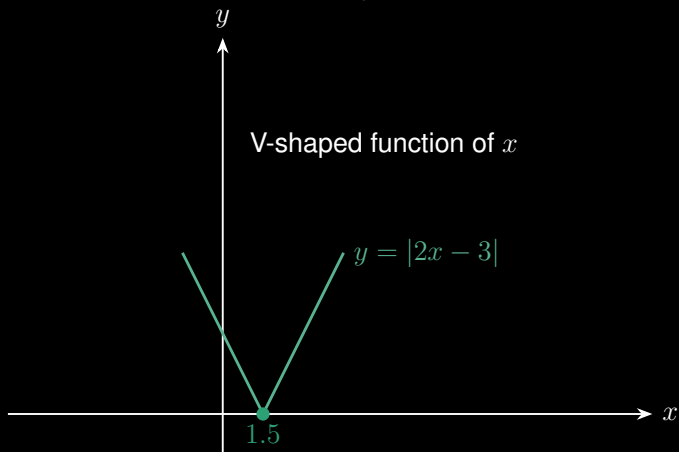
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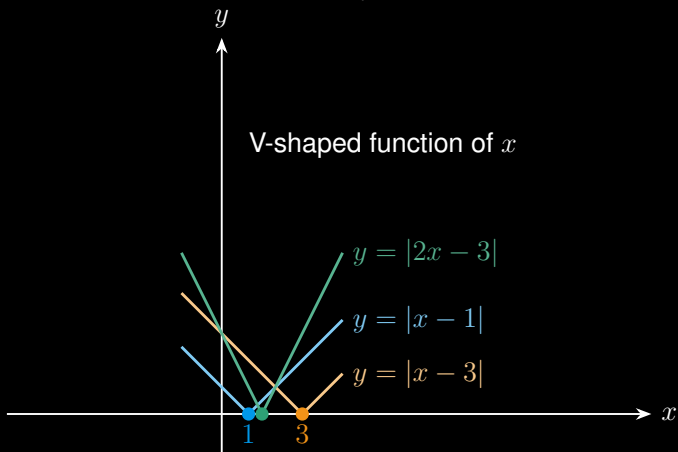
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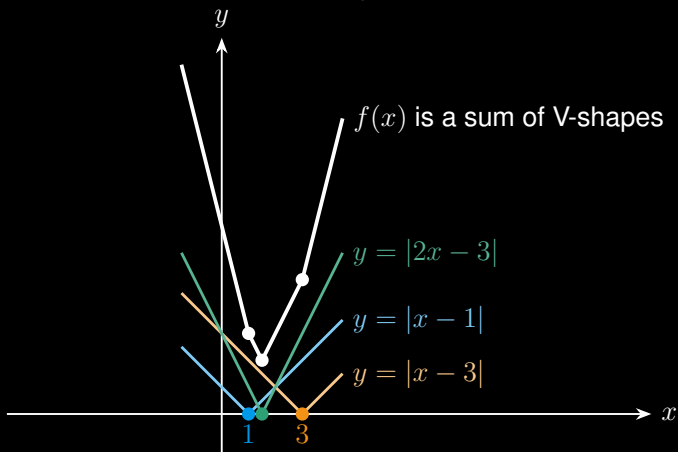
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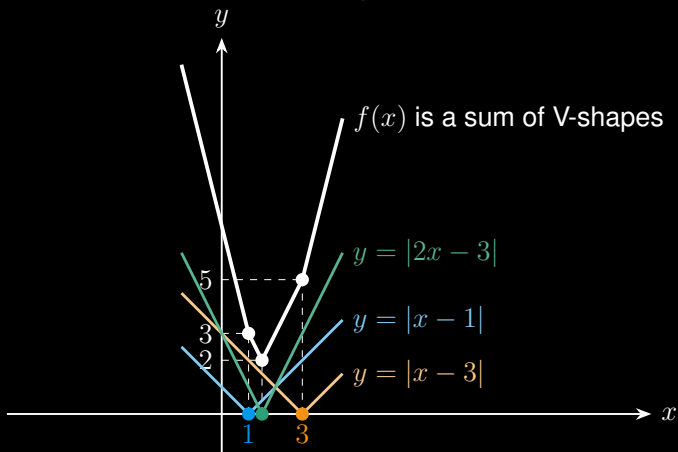
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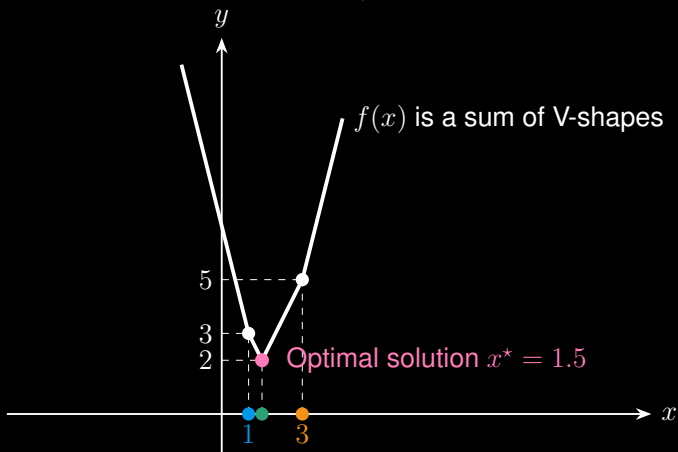
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Thanks for your attention!

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